

Product Requirement Document (PRD)

P.A.T.C.H. - Precision Assistant for Technical Context & Hardware - MVP

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1. Project Overview

P.A.T.C.H. (Precision Assistant for Technical Context & Hardware) is a web application that gives floor technicians immediate, source-referenced troubleshooting guidance for approved equipment documents. Teams upload equipment manuals, maintenance logs, and safety procedures. The system organizes them by machine, asset, and project, then uses AI to provide grounded answers with exact source references. The application will be built with Next.js, TypeScript, MongoDB, and a separate Python FastAPI AI microservice.

2. Problem Statement

Manufacturing firms maintain equipment manuals, maintenance logs, and safety procedures, but floor technicians cannot get an immediate, source-referenced fix during machine failures. Searching large documents and relying on individual experience increases downtime and can delay safe maintenance decisions.

The proposed workflow is:

Document Upload > Machine / Project Mapping > Search and Retrieval > AI Answer with Sources > Technician Review > Maintenance Log

3. Objectives

The MVP will:

- Accept equipment manuals, maintenance logs, and safety procedures.
- Store original documents and their metadata.
- Map documents to machine models, installed assets, and projects.
- Provide source-referenced answers to maintenance questions.
- Show exact document pages or sections used for an answer.
- Allow approved users to create and manage maintenance-log entries.
- Provide a dashboard for documents, machines, projects, and usage.

4. Target Users

- Floor Technicians: Find approved troubleshooting guidance during equipment failures.
- Maintenance Engineers: Upload documents, manage machine records, and review technical information.
- Safety Managers: Review and publish project-specific safety procedures.
- Administrators: Manage users, permissions, documents, machines, and projects.

5. Proposed Solution

5.1 Document Upload and Storage

Authorized users will upload original documents such as:

- Equipment manuals and OEM service documents
- Maintenance logs and completed work records
- Safety procedures and approved standard operating procedures
- Project-specific maintenance logs and custom safety procedures

Original files will be stored in their uploaded format. Each document version will record its title, type, revision, upload date, approval status, access permissions, and related machine or project.

5.2 Machine and Project Mapping

The system will maintain separate records for machine models and installed machine assets. A project can include multiple assets and may include additional project documents and approved safety procedures. This ensures answers are retrieved only from the sources relevant to the selected machine or project.

5.3 AI Integration

AI will only be used where normal search and document navigation are insufficient. The Next.js API layer will send approved AI requests to a separate Python FastAPI microservice, which performs retrieval and calls the AI provider. It will:

- Interpret a technician question and retrieve applicable document sections.
- Summarize a source-backed troubleshooting response.
- Present document title, version, page, and section references.
- Identify when available sources are incomplete, conflicting, or not applicable.
- Draft a maintenance-log note for technician review.

AI answers will not automatically control equipment or publish safety procedures. Approved human users remain responsible for maintenance and safety decisions.

5.4 Review Workflow

Users can:

- Select or search for a machine or project.
- Ask a maintenance question.
- Read the answer and open the cited document sources.
- Record whether the suggested information was useful.
- Create a maintenance-log entry for review.
- Review, edit, approve, and publish project-specific safety procedures.

5.5 MVP Scope

The MVP will not include direct machine control, IoT sensor integrations, automated work-order execution, live ERP/CMMS integrations, voice operation, offline mode, or automatic approval of AI-generated safety procedures. These features can be added after the MVP.

6. Key Features

Feature	Description
Document Upload	Upload manuals, maintenance logs, and safety procedures
Original File Storage	Preserve uploaded source documents and versions
Machine Management	Create machine-model and installed-asset records
Project Management	Group machines and project-specific documents
Document Mapping	Associate documents with machines and projects
AI Maintenance Assistant	Return source-referenced troubleshooting answers
Source Viewer	Open the cited document page or section
Maintenance Logs	Create and review maintenance-log entries
Safety Procedure Review	Review and publish approved procedures
Dashboard	Show documents, machines, projects, and usage statistics
Audit History	Store document versions, approvals, and user actions

7. Technology Stack

Technology	Purpose
Next.js	Frontend and backend application
TypeScript	Type-safe development
MongoDB	Store users, machines, projects, document metadata, logs, and audit history
Mongoose	MongoDB schema management
Python	AI microservice implementation language
FastAPI	AI microservice API for retrieval and answer generation
OpenAI API	Used by the FastAPI microservice for source-grounded answers
Zod	Server and form validation
Tailwind CSS	User-interface styling
Recharts	Basic dashboard charts
Auth.js	User login and role management
GitHub	Version control and collaboration

8. Functional Requirements

The system should be able to:

1. Allow users to log in based on their role.
2. Allow authorized users to upload original equipment documents.
3. Store document metadata, versions, approval status, and access permissions.
4. Create machine-model and installed-machine records.
5. Create projects and associate multiple machines with a project.
6. Map documents to the relevant machines and projects.
7. Process approved document text for search and retrieval.
8. Allow the Next.js API layer to communicate with the FastAPI AI microservice.
9. Allow technicians to ask a question for a selected machine or project.
10. Return an AI-generated answer with source citations.
11. Open the cited source document page or section.
12. Allow users to create and review maintenance-log entries.
13. Allow safety managers to approve and publish project-specific safety procedures.

9. Non-Functional Requirements

The application should be:

- Easy to understand and navigate on desktop and tablet screens.
- Secure and accessible only to logged-in users with the appropriate role.
- Able to preserve original document files and document-version history.
- Able to restrict answers to documents relevant to the selected machine or project.
- Reliable when the AI microservice or AI provider is unavailable by allowing document search and source viewing.
- Responsive enough to support active maintenance troubleshooting.
- Able to communicate securely between the Next.js API layer and the FastAPI AI microservice.
- Auditable, with document approval, procedure publication, and user-action history.
- Structured so additional document types, integrations, and retrieval rules can be added later.

10. Proposed Dashboard Flow

Dashboard 1 - Overview

- Total machines and active projects
- Total uploaded and approved documents
- Recent maintenance-log entries
- Recent questions and feedback
- Documents awaiting review

Dashboard 2 - Documents and Mapping

- Document upload form
- Document version and approval status
- Machine and project mappings
- Search indexing status
- Source-document preview

Dashboard 3 - Maintenance Assistant

- Machine or project selector
- Technician question input
- Source-referenced AI answer
- Cited document pages and sections
- Maintenance-log action

11. User Flow

Login > Select Machine or Project > Ask Maintenance Question > Retrieve Applicable Sources > Display Cited Answer > Technician Reviews > Create or Update Maintenance Log

12. Project Delivery

The complete project will be delivered as one finished submission. The work includes:

- Finalizing the database structure.
- Building authentication and user roles.
- Creating document upload and storage screens.
- Creating machine, asset, and project management screens.
- Storing document metadata, mappings, logs, and audit history in MongoDB.
- Implementing the Python FastAPI AI microservice.
- Connecting the Next.js API layer to the FastAPI microservice.
- Implementing document processing, search, and retrieval.
- Adding source-referenced AI maintenance answers.

- Building the source viewer and maintenance-log workflow.
- Building the safety-procedure review workflow.
- Creating the overview dashboard.
- Testing the complete workflow.
- Deploying the MVP.
- Completing the project documentation.

The final deliverables are:

- This Product Requirement Document (PRD), delivered in DOCX format
- Basic UX mock-up
- Working Next.js application
- MongoDB database
- GitHub repository
- Setup and deployment instructions

13. Success Criteria

The project will be successful when:

- An authorized user can upload an equipment document and retain its original file.
- Documents can be mapped to machine models, installed assets, and projects.
- A technician can select a machine or project and ask a maintenance question.
- The system returns a relevant answer with usable source references.
- Users can open the cited source document page or section.
- The system does not present an answer when no approved applicable source is available.
- Maintenance-log entries can be created and stored in MongoDB.
- Safety managers can review and publish project-specific safety procedures.
- The dashboard displays correct machine, document, project, and activity counts.
- The complete MVP workflow is deployed and demonstrated within two weeks.

14. Expected Outcome

The final MVP will give floor technicians one place to find approved, source-referenced maintenance guidance for the machine or project they are working on. It will preserve original documents, organize their relationship to assets and projects, and maintain audit history for documents, procedures, and maintenance activity.

The completed workflow will:

Document Upload > Machine / Project Mapping > Source Retrieval > AI Answer with Citations > Human Review > Maintenance Log

The MVP focuses on the main problem: helping technicians find the right source-backed information quickly during equipment failures, without replacing approved maintenance or safety decision-making.